INCH-POUND

A-A-58047A
30 July 2002
SUPERSEDING
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12 September 1995

COMMERCIAL ITEM DESCRIPTION
PROGRAMMABLE SYNTHESIZED SWEEP SIGNAL GENERATOR

10 MHz to 26.5 GHz

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE

1.1 This Commercial Item Description (CID), describes a solid state, general purpose, 10 MHz to 26.5 GHz, synthesized signal generator. This CID is meant as a minimum requirement for the synthesized signal generator in which only those manufacturers that meet or surpass the following requirements are qualified per this CID.

2. SALIENT CHARACTERISTICS

2.1 The equipment shall be capable of operation and be bid sample tested within the accuracies, limits, and specifications herein.

2.2 Equipment covered by this CID shall be commercially available equipment and may be modified to the extent necessary to meet the following description. The equipment shall be Class 3 in accordance with MIL-PRF-28800.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: WR-ALC/TILCC, 420 Second Street, Suite 100, Robins AFB, GA 31098-1640.

AMSC N/A FSC 6625

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2.3 Order of Precedence. In the event of a conflict between the text of this CID and the references cited herein (except for associated detail specifications, specification sheets or MS standards) the text of this CID shall take precedence. Nothing in this CID, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

2.4 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein.

2.5 Electrical Power Sources. The equipment shall operate from nominal commercial, military, and shipboard power sources of 110/220 volts AC (±10%) at line frequencies of 50 Hz to 60 Hz. Power consumption shall not exceed 400 VA. Transient state conditions shall meet MIL-PRF-28800 requirements for the classification specified herein.

2.6 Temperature Requirements.

2.6.1 Operating Temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of 0°C to +50°C.

2.6.2 Nonoperating Temperature. The equipment shall meet its performance and accuracy requirements in an nonoperating environment of -40°C to +71°C.

2.7 Mean Time Between Failures (MTBF). The equipment Mean Time Between Failures (MTBF) shall be 2,000 hours. The lower test MTBF shall be 2,000 hours and the upper test MTBF shall be 4,000 hours.

2.8 Pre-conditioning Requirement. Each deliverable equipment shall withstand the environmental screening effects of burn in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

2.9 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater based on an operating time of 2000 hours.

2.10 Performance Characteristics. The synthesized signal generator shall be ASCII controllable language through an external IEEE 488 bus configuration per MIL-PRF-28800.

2.10.1 Frequency. The minimum frequency range of the synthesized signal generator shall be 10 MHz to 26.5 GHz.

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2.10.1.1 Frequency Resolution:

1 Hz, 0.01 to 7.0 GHz
2 Hz, 7.0 to 13.5 GHz
3 Hz, 13.5 to 20.0 GHz
4 Hz, 20.0 to 26.5 GHz

2.10.1.2 Accuracy. The frequency accuracy shall be based on the internal reference oscillator aging rate which shall be 1 x 10^{-9} per day after 10 days of continuous operation.

2.10.1.3 Switching Speed. Signal Generator will switch frequencies in 50 milliseconds maximum.

2.10.1.4 Sweep Time. The minimum sweep time shall be at least 45 milliseconds.

2.10.2 Spectral Purity.

2.10.2.1 Harmonics. The synthesized signal generator harmonics shall be -35 dBc or greater over the entire frequency range.

2.10.2.2 Sub-Harmonics. The sub-harmonics shall be -25 dBc or greater to 18 GHz and -20 dBc or greater 18 to 26 GHz.

2.10.2.3 Spurious Non-Harmonics. The spurious non-harmonics shall be at least -50 dBc over entire frequency spectrum.

2.10.2.4 Absolute SSB Phase Noise. The absolute single sideband (SSB) phase noise at 26 GHz carrier at 100 KHz offset shall be at least -90 dBc per Hz.

2.10.3 Output. The minimum output level range into 50 ohm nominal shall be 8 dBm to -99 dBm for frequencies up to 20 GHz, and 1 dBm to -99 dBm for frequencies from 20 to 26.5 GHz.

2.10.3.1 Output Resolution. The synthesized signal generator output resolution shall be 0.1 dB.

2.10.3.2 Output Protection. The synthesized signal generator shall have reverse power protection to +20 dBm.

2.10.3.3 Output Connector. The RF output connector shall be APC-3.5 female.

2.10.3.4 Output Accuracy. The output accuracy shall be ±2 dB.

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2.10.4 Sweep Mode. The signal generator shall provide analog and/or digital sweep capability.

2.10.4.1 Center Frequency/Sweep Width (ΔF).

2.10.4.1.1 Range: 10 MHz to 26.5 GHz nominal (center frequency) 10 MHz to 26.5 GHz nominal (sweep width).

2.10.4.1.2 Resolution: Approximately 0.1% of sweep width (ΔF).

2.10.4.2 Start/Stop Frequency

2.10.4.2.1 Range: 10 MHz to 26.499 GHz nominal (Start), 10.5 MHz to 26.5 GHz nominal (Stop).

2.10.4.2.2 Resolution: Approximately 0.1% of sweep width (ΔF).

2.10.4.3 Markers. The signal generator shall have at least 5 independent markers.

2.10.5 Pulse Modulation. The on/off ratio for the pulse modulation shall be 80 dB or greater.

2.10.5.1 Rise and Fall Time. The rise and fall time between 10% and 90% of the pulse shall be 25 nanoseconds or less.

2.10.5.2 Repetition Rate. The minimum repetition rate for a carrier frequency of 10 MHz to 26.5 GHz shall be 100 Hz to 1 MHz.

2.10.5.3 Pulse Width. The minimum pulse width shall be at least 100 nanoseconds.

2.10.6 Amplitude Modulation.

2.10.6.1 AM Depth: 90% Typical. Actual available depth will be greater than this in many cases and is determined by the difference of the maximum leveled power available at frequency of interest and -30 dBm.

2.10.6.2 AM Sensitivity: (at a 1 KHz rate and 30% depth) 100%/V +5%. AM depth is linearly controlled by varying input level between 0 and +1V peak.

2.10.6.3 AM Bandwidth (30% Depth, PULSE off): 3 dB point 10 KHz.

2.10.7 Frequency Modulation.

2.10.7.1 Modulation Rates: 50 KHz to 1 MHz.

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2.10.7.2 Peak Deviation: the lesser of 10 MHz or:

Band 1 : 5 X Mod Rate
Band 2 : 10 X Mod Rate
Band 3 : 15 X Mod Rate
Band 4 : 20 X Mod Rate

2.10.7.3 Deviation Accuracy: ±10% at 100 kHz Rate.

2.10.8 Dimensions and Weight. The size and weight of the equipment shall be consistent with current commercial capabilities. The nominal dimensions for the synthesized signal generator should be approximately 8 inches high, 17 inches wide, and 24 inches deep. The weight including all accessories (excluding manual), shall not exceed 80 lbs.

3. REGULATORY REQUIREMENTS

3.1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS

4.1 Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.2 The contracting officer has the option of accepting or rejecting the product.

5. PACKAGING

5.1 Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

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6. NOTES

6.1 Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this Commercial Item Description are met. If a product is manufactured to metric dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

6.2 The contracting officer has the option of accepting or rejecting the product.

6.3 MIL-PRF-28800 is available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.

6.4 Ordering data. Procurement documents must specify the title, number, and date of this CID, the issue of the DoDISS to be used in the solicitation, and preservation and packaging requirements.

MILITARY INTERESTS

Preparing Activity
Air Force – 84

Agent
Air Force – 99

Project No 6625-0909

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